

Pamantasan ng Lungsod ng Maynila
Department of Electronics Engineering

ENTERPRISE NETWORK ARCHITECTURE

*Design and Simulated Implementation of a
Resilient Multi-Floor Campus Network for the
New CAAP Building, Pasay City*

A Project Report
Submitted in Partial Fulfillment of the Requirements for the Course in
Enterprise Network Architecture

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Executive Summary

This project presents the design and simulated implementation of a resilient, multi-floor enterprise campus network for the New CAAP Building located along NAIA Road in Pasay City. The design serves five occupied floors (Ground Floor through Fifth Floor) and is implemented in Cisco Packet Tracer using a representative subset of the full topology in accordance with the device-reduction strategy permitted by the course brief.

The network applies the Cisco hierarchical internetworking model in a collapsed-core configuration. A single edge router (EDGE-RTR) terminates two upstream ISP links and performs Network Address Translation. Two collapsed-core multilayer switches (CORE-1 and CORE-2) host all Switched Virtual Interfaces, run Hot Standby Router Protocol version 2 in an odd/even load-split arrangement, and peer with the edge router over Open Shortest Path First. An inter-core routed EtherChannel (Port-channel 1) provides high-bandwidth east-west connectivity between the cores. Access switches are deployed in each floor's distribution frame, with two trunk uplinks per access switch terminating on CORE-1 and CORE-2 respectively.

Core network services — Dynamic Host Configuration Protocol, Domain Name System, Network Time Protocol, and Syslog — are deployed on two centralized servers in VLAN 30, with DHCP relay configured on every user-facing SVI to forward client requests to the DHCP server.

Parameter	Value
Site	New CAAP Building, NAIA Road, Pasay City
Floors served	5 (Ground Floor, F2, F3, F4, F5)
Architecture	Collapsed Core, 3-tier conceptual / 2-tier physical
Edge routers	1 (EDGE-RTR) with dual-ISP failover
Collapsed-core switches	2 (CORE-1, CORE-2) with HSRPv2 odd/even split
Inter-core link	Layer-3 EtherChannel (Port-channel 1, LACP)
Access switches	Representative subset (1 per floor in PT)
VLANs (active)	15 user/service VLANs + 1 native (16 total)
Routing protocol	OSPFv2 single Area 0
First-hop redundancy	HSRPv2 (ODD on CORE-1, EVEN on CORE-2)
DHCP server / NTP source host	Server-2 (10.10.30.20), Packet Tracer DHCP service
DNS / Syslog	Server-1 (10.10.30.10)
WAN ISPs	ISP-A (primary), ISP-B (floating static, AD 5)
NAT	PAT overload on EDGE-RTR (pool 203.0.113.200/29)

1. Introduction

1.1 Project Background and Site

The chosen site is the New CAAP Building along NAIA Road, Pasay City — a publicly documented multi-storey office building whose architectural drawings (floor plans, front elevation) ground the physical-layer decisions (cable distances, distribution frame placement, riser routing).

The tenant is treated as a hybrid CAAP-inspired enterprise: real building geometry and pre-provisioned voice/data infrastructure are used, but the operational complexity of the actual Civil Aviation Authority is not replicated.

1.2 Objectives

The objectives of this project are:

- To design a hierarchical enterprise network for five occupied floors of the New CAAP Building, providing wired and wireless connectivity, IP telephony, network printing, and centralized services to all users.
- To provide redundancy at the first-hop layer through dual collapsed-core switches running HSRPv2, with traffic load-distributed across both cores using an odd/even VLAN split.
- To provide redundancy at the WAN edge through a single edge router with two ISP uplinks configured as a primary route and a floating static backup.
- To deploy dynamic routing using OSPF Area 0 across all routed interfaces of the internal infrastructure.
- To deliver core services — DHCP, DNS, NTP, and Syslog — centrally and verify their correct operation across all VLANs.
- To document the design with sufficient depth to support an oral defense, including a defense briefing of likely questions with prepared answers.

1.3 Scope and Limitations

Implementation is in Cisco Packet Tracer. The simulation reproduces L2/L3 protocol behavior (802.1Q, Rapid PVST+, HSRPv2, OSPFv2, LACP, NAT, DHCP relay) but not physical-layer phenomena.

Explicit limitations of the present iteration:

- Device-count reduction is applied. The full deployment would include four access switches per floor; the Packet Tracer model includes one representative access switch per floor (five total), and the remaining fifteen are documented but not instantiated.
- Server-side functions (DHCP service, DNS resolver, Syslog collector, NTP master) are demonstrated on two centralized servers rather than a clustered or high-availability arrangement.
- Layer-2 hardening features such as DHCP Snooping, Dynamic ARP Inspection, and IP Source Guard are not configured in this iteration; recommendations for their inclusion appear in Section 9.

- Port-security with sticky MAC learning was evaluated and intentionally omitted from the demonstration to avoid simulation-specific recovery issues during testing; the design rationale and a production-ready template are provided in Section 9.
- Authentication, authorization, and accounting (AAA) using an external RADIUS or TACACS+ server is not implemented; local user accounts secure all management access.
- Quality of Service (QoS) policies for voice traffic are not configured. The design provides per-floor VLAN separation between voice and data but does not enforce DSCP marking or priority queuing.
- IPv6 is not deployed in this iteration.

1.4 Assumptions

- Each ISP provides a single point-to-point /30 transit link with a known upstream next-hop address. RFC 5737 documentation prefixes are used in place of real public address space for the academic exercise.
- All inter-device links operate at speeds sufficient to avoid congestion at the modeled user count. No oversubscription analysis is performed.
- Endpoints requiring voice service (IP phones) negotiate the voice VLAN tag using CDP or LLDP-MED, as is standard for Cisco IP phones with an attached PC.
- The DHCP server function is hosted on a centralized Packet Tracer server. All user-facing SVIs on both cores are configured with ip helper-address to forward DHCP broadcasts to the server.
- The building's pre-provisioned voice-and-data infrastructure — distribution frames, server racks in each AUX ROOM, and the vertical riser system — is available for tenant use.

2. Requirements

2.1 Functional Requirements

- Provide IPv4 connectivity to all wired and wireless endpoints across five floors.
- Separate broadcast domains for data and voice traffic on each floor, in line with the standard practice for IP telephony deployments.
- Provide a single, consistent default gateway per user VLAN that survives the failure of either collapsed-core switch.
- Provide internet access to all internal endpoints through Network Address Translation performed at the edge.
- Provide centralized DHCP services with per-VLAN scopes for automatic address assignment.
- Provide centralized DNS resolution.
- Provide synchronized time across all infrastructure devices through NTP.
- Provide centralized log aggregation through Syslog.
- Survive any single edge-link failure, core-switch failure, or access-uplink failure without isolating any user VLAN.

2.2 Non-Functional Requirements

Redundancy and Availability

Redundancy is enforced at three layers. At the WAN edge, two upstream ISPs are used: ISP-A as the primary path and ISP-B as a floating static backup with administrative distance five. At the collapsed core, two multilayer switches run HSRPv2 with an odd/even VLAN load split, so that half the user VLANs converge through CORE-1 and the other half through CORE-2 under normal operation, but either core can serve all VLANs in a failure scenario. At the access layer, each access switch is dual-homed to both cores via 802.1Q trunks, with Rapid PVST+ aligned to the HSRP active core on a per-VLAN basis.

Scalability

The VLAN numbering scheme uses the second digit of two-digit VLAN identifiers to encode the floor (VLAN 11 = Data, Ground Floor; VLAN 15 = Data, Fifth Floor) and the first digit to encode the traffic category (VLAN 1X = Data, VLAN 2X = Voice). Adding a sixth floor would consume VLAN 16 (DATA_F6) and VLAN 26 (VOICE_F6), the corresponding /24 subnets, and a new pair of HSRP groups. No renumbering of existing infrastructure would be required.

Manageability

All infrastructure devices carry an SVI in VLAN 40 (MGMT) with an IP within 10.10.40.0/26. SSH version 2 is enabled on every internal device, with VTY access controlled by access-list 1 to permit only sources within 10.10.40.0/26. NTP and Syslog are centralized so that time-correlated log analysis across devices is straightforward.

Security (Baseline)

Baseline measures applied in this iteration: SSH-only management access; VTY ACL restricting SSH to the MGMT subnet; service password-encryption enabled on every internal device; native VLAN on trunks set to VLAN 100 (a routed-nowhere blackhole) rather than the default VLAN 1; switchport nonegotiate on all trunks to prevent DTP-based attacks; unused range f0/22 to f0/24 on cores administratively shut. Further hardening is identified in Section 9 as future work.

2.3 Endpoint Inventory

The deployment serves five floors with the following endpoint categories. The Packet Tracer model represents the design at a reduced scale (Section 6.1) sufficient for demonstration and defense.

Floor	Endpoint Categories Present	Server Presence	Wireless
Ground Floor (GF)	Reception PC + phone, printers, servers, IT operations	Server-1, Server-2 (VLAN 30)	Corporate + Guest APs
Second Floor (F2)	User PCs + phones, printers, laptops	None	Corporate + Guest APs
Third Floor (F3)	User PCs + phones, printers, laptops	None	Corporate + Guest APs
Fourth Floor (F4)	User PCs + phones, printers, laptops	None	Corporate + Guest APs
Fifth Floor (F5)	User PCs + phones, printers, laptops	None	Corporate + Guest APs

3. Network Design

3.1 Design Philosophy

The design applies the Cisco hierarchical model in a collapsed-core arrangement: Core and Distribution roles merge into one tier of multilayer switches. This is appropriate for a single-building, single-distribution-block deployment of this scale.

The three logical layers in this design are:

- Access layer: a single Layer-2 switch per floor (ACC-GF-1 through ACC-F5-1 in the reduced Packet Tracer model; one of four per floor in the full design). Each access switch applies port-level policy (VLAN assignment, voice VLAN, PortFast) and connects end users to the network.
- Collapsed core: two multilayer switches, CORE-1 and CORE-2. Host SVIs for every user, voice, and service VLAN; run HSRPv2 for first-hop redundancy; peer with the edge router over OSPF; provide inter-VLAN routing.
- Edge: a single router (EDGE-RTR) terminating two upstream ISP links, performing NAT overload for outbound traffic, and originating a default route into the OSPF domain.

3.2 Logical Topology

Access switches trunk all locally relevant VLANs up to both cores. The cores interconnect via a routed Layer-3 EtherChannel (Port-channel 1) and each terminate an independent routed link to EDGE-RTR. EDGE-RTR has two fiber WAN uplinks to ISP-A (primary) and ISP-B (backup).

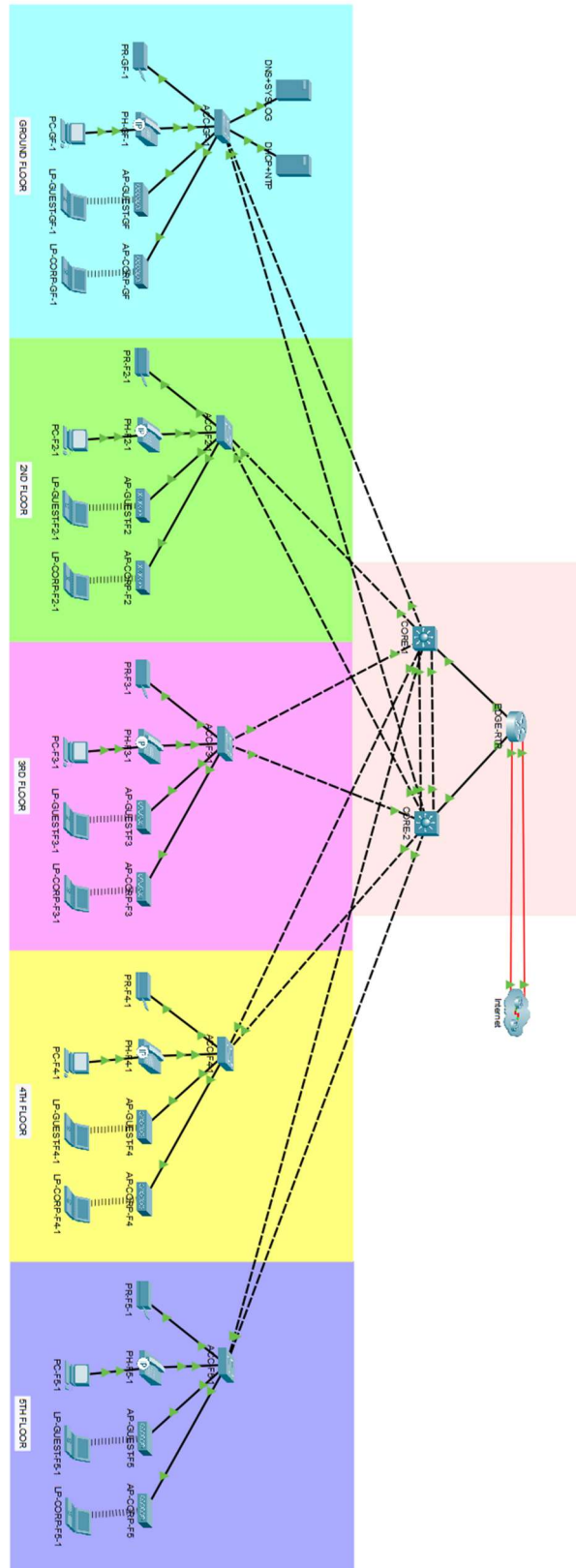


Figure 1: Logical Topology Diagram (PT screenshot)

3.3 Physical Topology and Distribution Frames

Physical layout follows the building's pre-provisioned voice-and-data infrastructure:

- The Main Distribution Frame (MDF) is located in the Ground Floor SERVER RACK within the AUX ROOM near gridline D, adjacent to the building's vertical riser shaft. The MDF houses EDGE-RTR, CORE-1, CORE-2, the four GF access switches (in the full design), Server-1, and Server-2.
- An Intermediate Distribution Frame (IDF) is located in each upper floor's SERVER RACK within that floor's AUX ROOM. Each IDF houses up to four access switches in the full design (one per floor in the reduced Packet Tracer model).
- The IDFs on all floors are vertically aligned with the MDF at gridline D, so that the vertical fiber riser is a straight run from the GF MDF to the topmost IDF (F5).

Cable types used in the design:

Segment	Media	Rationale
End host (PC/IP phone/printer) to access switch	Cat6 UTP	Within 90 m channel limit; standard horizontal cabling
Access switch (IDF) to collapsed core (MDF) riser	OM4 multimode fiber, 10GBASE-SR	Future-proof, EMI-immune, industry standard for risers
CORE to EDGE-RTR and CORE-1 to CORE-2 (within MDF)	Copper Ethernet (Cat6) or fiber patch	Short intra-rack runs; either media class is acceptable
EDGE-RTR to ISP-A and ISP-B (WAN handoff)	Single-mode fiber via SFP (GLC-LH-SMD)	Realistic WAN handoff to provider

3.4 Cabling Distance and Riser Capacity

Worst-case horizontal Cat6 runs from a floor's IDF to the farthest user outlet are conservatively estimated at 35 to 45 m, well within the TIA/EIA-568 90 m horizontal channel limit. Vertical fiber riser runs from the GF MDF to each upper-floor IDF, based on the building's front elevation:

IDF Location	Approximate Vertical Run (MDF to IDF)
F2 IDF	~5.0 m
F3 IDF	~9.0 m
F4 IDF	~13.0 m
F5 IDF	~17.0 m

Each floor's riser is a home run to the MDF (no daisy-chaining between floors), so a riser fault on one floor does not affect any other floor. Each access switch consumes two strand-pairs (one to CORE-1, one to CORE-2). For four access switches per floor in the full design, this is 8 fiber pairs (16 strands) per floor; a 24-strand OM4 bundle per floor leaves 8 spare strands for repair or future expansion.

3.5 Redundancy Strategy

Edge-Layer Redundancy

EDGE-RTR terminates two ISP links on physically distinct interfaces. ISP-A is the primary path; the default route through ISP-A has administrative distance 1 (the IOS default for static routes). ISP-B is a floating static with administrative distance 5; it remains inactive in the routing table while ISP-A is reachable, but takes over automatically when the ISP-A next hop becomes unreachable through interface-state detection. The default route is then originated into OSPF using default-information originate, so the cores and downstream devices always have a default path through whichever ISP is currently active.

Core-Layer Redundancy

CORE-1 and CORE-2 are interconnected by a Layer-3 routed EtherChannel (Port-channel 1) using LACP. The bundle aggregates two GigabitEthernet members (G0/1 and G0/2 on each core) and is addressed with the /30 subnet 10.99.0.8/30. Bundling at Layer 3 keeps the inter-core link out of the Spanning Tree topology and allows OSPF to maintain a coherent view of the routed core. If one member link fails, traffic immediately rebalances onto the remaining member without recomputing routes.

First-Hop Redundancy

HSRPv2 is configured on every user-facing and service-facing SVI on both cores. Each VLAN's SVI on CORE-1 and CORE-2 shares a common Virtual IP (VIP), which serves as the default gateway for all hosts in that VLAN. The HSRP load-split strategy is detailed in Section 5.2.

3.6 IP Addressing Plan

The IP addressing plan uses RFC 1918 private space internally, allocated by purpose using the second octet:

Range	Category	Description
10.1.X.0/24	Data	Per-floor user data subnets (X = 11–15)
10.2.X.0/24	Voice	Per-floor IP-telephony subnets (X = 21–25)
10.10.30.0/26	Servers	Centralized server farm in MDF (VLAN 30)
10.10.40.0/26	Management	Switch and router SVI addresses (VLAN 40)
10.10.50.0/27	Printers	Network printers (VLAN 50)
10.20.60.0/23	Wi-Fi Corporate	Corporate WLAN clients (VLAN 60)
10.20.70.0/24	Wi-Fi Guest	Guest WLAN clients (VLAN 70)
10.99.0.0/30	Transit	EDGE-RTR ↔ CORE-1
10.99.0.4/30	Transit	EDGE-RTR ↔ CORE-2
10.99.0.8/30	Transit	CORE-1 ↔ CORE-2 (Port-channel 1)
10.99.0.51/32, .52/32, .53/32	Loopbacks	EDGE-RTR, CORE-1, CORE-2 (OSPF router-id, mgmt)
203.0.113.0/30, .4/30	WAN	EDGE-RTR to ISP-A and ISP-B (RFC 5737 documentation block)

203.0.113.200/29	NAT pool	Outside global addresses used for PAT overload
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Within each user subnet, low addresses follow a uniform convention: .1 is the HSRP VIP (default gateway), .2 is the SVI on CORE-1, .3 is the SVI on CORE-2, and .4 through .10 are reserved for static infrastructure or future allocation. DHCP starts issuing at .11.

4. VLAN and Layer-2 Design

4.1 VLAN Numbering Convention

The VLAN numbering is self-documenting:

VLAN ID	Name	Subnet	Mask	Gateway (VIP)
11	DATA_GF	10.1.11.0	/24	10.1.11.1
12	DATA_F2	10.1.12.0	/24	10.1.12.1
13	DATA_F3	10.1.13.0	/24	10.1.13.1
14	DATA_F4	10.1.14.0	/24	10.1.14.1
15	DATA_F5	10.1.15.0	/24	10.1.15.1
21	VOICE_GF	10.2.21.0	/24	10.2.21.1
22	VOICE_F2	10.2.22.0	/24	10.2.22.1
23	VOICE_F3	10.2.23.0	/24	10.2.23.1
24	VOICE_F4	10.2.24.0	/24	10.2.24.1
25	VOICE_F5	10.2.25.0	/24	10.2.25.1
30	SERVERS	10.10.30.0	/26	10.10.30.1
40	MGMT	10.10.40.0	/26	10.10.40.1
50	PRINTERS	10.10.50.0	/27	10.10.50.1
60	WIFI_CORP	10.20.60.0	/23	10.20.60.1
70	WIFI_GUEST	10.20.70.0	/24	10.20.70.1
100	NATIVE	(not routed)	—	(none)

VLAN 1 (the Cisco default VLAN) is not used for any production traffic. VLAN 100 (NATIVE) is the configured native VLAN on every 802.1Q trunk and has no SVI on either core; untagged frames received on a trunk are therefore associated with VLAN 100 and silently dropped without routing.

4.2 VTP Configuration

VTP is operated in transparent mode (version 2), with the domain name data.com on every switch. Transparent mode is selected to avoid the operational hazards of VTP server/client propagation: in transparent mode, each switch maintains its own VLAN database, no switch can accidentally erase the VLAN list across the campus, and configuration intent remains explicit on each device. The trade-off is that VLANs must be defined on every switch where they appear, which is acceptable at this scale.

4.3 Trunking and Allowed-VLAN Pruning

Every inter-switch link is configured as an 802.1Q trunk with the following uniform properties:

- switchport mode trunk and switchport nonegotiate — eliminates DTP and prevents dynamic trunk negotiation attacks.
- switchport trunk encapsulation dot1q — explicit, even where it is the only available encapsulation.
- switchport trunk native vlan 100 — moves the native off the default VLAN 1 to mitigate double-tag VLAN-hopping attacks.
- switchport trunk allowed vlan <pruned list> — only VLANs that legitimately need to traverse a given trunk are permitted on it.

Allowed-VLAN pruning is enforced on a per-floor basis. The trunks from each upper-floor access switch carry only that floor's data and voice VLANs plus shared infrastructure VLANs:

Trunk	Allowed VLANs
ACC-GF-1 ↔ Cores	11, 21, 30, 40, 50, 60, 70, 100
ACC-F2-1 ↔ Cores	12, 22, 40, 50, 60, 70, 100
ACC-F3-1 ↔ Cores	13, 23, 40, 50, 60, 70, 100
ACC-F4-1 ↔ Cores	14, 24, 40, 50, 60, 70, 100
ACC-F5-1 ↔ Cores	15, 25, 40, 50, 60, 70, 100

VLAN 30 (SERVERS) is allowed only on the Ground Floor access trunks because the servers physically reside in the GF MDF; permitting it on upper-floor trunks would extend the server broadcast domain unnecessarily and increase the attack surface.

4.4 Spanning Tree (Rapid PVST+)

Rapid PVST+ is the chosen spanning tree mode on every switch. STP root-bridge assignment is explicitly aligned to the HSRP active core on a per-VLAN basis, so that the Layer-2 forwarding path follows the Layer-3 active gateway and avoids suboptimal traffic patterns:

- CORE-1 is configured with spanning-tree vlan 11,13,15,21,23,25,30,40,50,60,70,100 priority 0 — root bridge for all ODD-numbered user VLANs and the shared service VLANs, matching the HSRP active assignment.
- CORE-2 is configured with spanning-tree vlan 12,14,22,24 priority 0 — root bridge for the EVEN user VLANs where it is HSRP active.
- Each core is also configured with priority 4096 on the VLANs where the other core is root, ensuring deterministic secondary-root behavior in failure scenarios.

4.5 Voice VLAN Integration

Cisco IP phones share a single physical access port with a desktop PC by exploiting the voice VLAN feature. The switch advertises the voice VLAN tag to the attached IP phone using CDP (or LLDP-MED), so the phone tags its own traffic with the voice VLAN ID while the PC behind the phone remains untagged. The result is one cable carrying two VLANs:

```
interface FastEthernet0/10
description USER ACCESS - DATA_GF + VOICE_GF
switchport mode access
```



```
switchport access vlan 11      ! Data
switchport voice vlan 21      ! Voice
spanning-tree portfast
no shutdown
```

This pattern is replicated across all user-facing ports on every floor, with the data and voice VLAN values substituted for the appropriate floor.

5. Layer-3 Design

5.1 Inter-VLAN Routing

All inter-VLAN routing happens on CORE-1 and CORE-2. Every VLAN has an SVI on both cores. EDGE-RTR routes only internet-bound traffic.

5.2 First-Hop Redundancy: HSRPv2 with Odd/Even Load Split

HSRPv2 is configured on every user-facing and service-facing SVI on both cores. Version 2 supports group IDs > 255, IPv6, and a distinct virtual MAC range, leaving room for future expansion.

- It supports group identifiers greater than 255, which keeps the door open for future numbering schemes.
- It uses a different virtual MAC range (0000.0C9F.FXXX) and a different multicast address (224.0.0.102), which is less likely to collide with other multicast traffic.
- It supports IPv6, in case the design is extended.

Load-Split Strategy

Active responsibility for each VLAN is distributed across the two cores using an odd/even split on the user VLAN identifiers. CORE-1 is HSRP Active for ODD-numbered user VLANs (11, 13, 15, 21, 23, 25) and for all shared service VLANs (30, 40, 50, 60, 70). CORE-2 is HSRP Active for EVEN-numbered user VLANs (12, 14, 22, 24).

This produces a deterministic, easily-verified load distribution: half the user floors enter the routing domain through CORE-1 and the other half through CORE-2 under normal conditions. The shared service VLANs (Servers, Management, Printers, both Wi-Fi VLANs) follow CORE-1. Either core can serve all VLANs in a failure scenario because every SVI exists on both cores; only the priority differs.

Priority, Preemption, and the Virtual IP

For each VLAN, the Active core is configured with priority 110, preempt enabled, and the VIP (.1 in the subnet). The Standby core uses the IOS-default priority of 100 and the same VIP. Because preempt is enabled and the Active core has the higher priority, the original Active reclaims its role after any recovery event. A representative SVI configuration on CORE-1 (active for VLAN 11) is:

```
interface Vlan11
  description DATA_GF - Core-1 active (ODD)
  ip address 10.1.11.2 255.255.255.0
  ip helper-address 10.10.30.20
  standby version 2
  standby 11 ip 10.1.11.1
  standby 11 priority 110
  standby 11 preempt
  no shutdown
```

The corresponding SVI on CORE-2 (standby for VLAN 11) is identical except that the SVI IP is 10.1.11.3 and the priority lines are absent (defaulting to 100, no preempt).

5.3 OSPFv2 Single-Area Backbone

OSPFv2 is the IGP. Single Area 0 — multi-area would add complexity for no benefit at this scale (3 speakers, 4 adjacencies).

Router-IDs from Loopbacks

Each OSPF speaker is configured with a Loopback0 interface bearing a /32 address within 10.99.0.0/24, and the OSPF router-id is anchored to that address:

Device	Loopback0 / OSPF Router-ID
EDGE-RTR	10.99.0.51
CORE-1	10.99.0.52
CORE-2	10.99.0.53

Passive Interfaces

All SVIs on the cores are declared passive in the OSPF process. This is mandatory because user-facing SVIs face hosts, not OSPF neighbors; sending OSPF hellos on a user VLAN would be wasteful and expose the protocol to untrusted hosts. The transit interfaces (the routed ports to the edge and Port-channel 1 between cores) are not passive, so neighbor relationships form there.

Point-to-Point Network Type on Transit Links

Each routed transit interface is configured with `ip ospf network point-to-point`. The default broadcast network type on an Ethernet interface would force a Designated Router election, which has no value on a two-router link and slows adjacency convergence. The point-to-point type skips DR/BDR election and advertises the configured /30 subnet directly.

5.4 Internet Edge: NAT and Default Routing

EDGE-RTR performs two functions in addition to OSPF: NAT overload and default-route origination.

NAT (PAT Overload)

Internal RFC 1918 addresses (10.1.0.0/16, 10.2.0.0/16, 10.10.0.0/16, 10.20.0.0/16) are translated to a small pool of outside global addresses (203.0.113.200 to 203.0.113.207, an /29) using port-address translation (overload). The inside interfaces are the two routed ports to CORE-1 and CORE-2; the outside interfaces are the two WAN-facing ports to ISP-A and ISP-B.

Dual-ISP Failover

Two static default routes are configured:

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1      ! Primary via ISP-A (AD 1)
ip route 0.0.0.0 0.0.0.0 203.0.113.5 5    ! Floating backup via ISP-B (AD 5)
```

The ISP-A route is preferred because its administrative distance (1, IOS default) is lower than the ISP-B route's (5). If the ISP-A next hop becomes unreachable (the route is removed from the RIB when the outgoing interface goes down), the floating static through ISP-B is installed. EDGE-RTR

then injects the default into OSPF using default-information originate, so the cores and downstream switches always learn a default route that points to whichever path is currently active.

Return Routing on the ISPs

Each simulated ISP carries a static return route covering the NAT pool, so that replies to the translated outside-global addresses route back to EDGE-RTR:

```
! On ISP-A and ISP-B
ip route 203.0.113.200 255.255.255.248 <EDGE-RTR-side IP>
```

5.5 Transit Link Inventory

Subnet	Purpose	Side A	Side B
10.99.0.0/30	EDGE-RTR ↔ CORE-1	EDGE-RTR G0/0: .1	CORE-1 F0/1: .2
10.99.0.4/30	EDGE-RTR ↔ CORE-2	EDGE-RTR G0/1: .5	CORE-2 F0/1: .6
10.99.0.8/30	CORE-1 ↔ CORE-2 (Po1, LACP)	CORE-1 Po1: .9	CORE-2 Po1: .10
203.0.113.0/30	ISP-A ↔ EDGE-RTR	ISP-A G0/0/0: .1	EDGE-RTR G0/0/0: .2
203.0.113.4/30	ISP-B ↔ EDGE-RTR	ISP-B G0/0/0: .5	EDGE-RTR G0/1/0: .6

6. Services

6.1 DHCP

DHCP service is hosted on Server-2 (10.10.30.20) using the Packet Tracer server's built-in DHCP service. Per-VLAN scopes are defined on the server (the complete scope reference is in Appendix A). Every user-facing SVI on both cores carries an ip helper-address 10.10.30.20 directive, so that DHCP DISCOVER broadcasts originating in any user VLAN are converted to unicasts and forwarded to Server-2 with the appropriate giaddr (gateway IP address) field. Server-2 uses the giaddr to select the matching scope.

Scope Design Choices

- The default-router for every scope is the HSRP VIP (.1), not the SVI IP of either physical core. This ensures the gateway address learned by clients survives HSRP failover transparently.
- The Start IP for each user scope is .11. Addresses .1 through .10 are reserved for the VIP, the two SVI IPs, and other static infrastructure.
- Voice scopes include Option 150 (TFTP server) pointing at 10.10.30.10 for IP phone bootstrap configuration retrieval.
- The Guest Wi-Fi scope uses 8.8.8.8 as its DNS server rather than the internal resolver, isolating guest DNS queries from the internal namespace.
- Server-1, Server-2, and all switch/router management interfaces use static addressing; they do not consume DHCP leases.

DHCP Relay Configuration

DHCP relay is configured on both cores (Active and Standby) for every user VLAN. Configuring helper-address on the Standby core as well as the Active is critical: if helper-address were configured only on the Active and the Active failed, the new Active (formerly Standby) would have no helper and DHCP would silently stop working for new leases on that VLAN until manual intervention. The current configuration eliminates that failure mode.

6.2 DNS

Domain Name System resolution is provided by a centralized DNS server running on Server-1 (10.10.30.10). Endpoints receive the server address via DHCP option 6 (dns-server) and direct all name-resolution queries to it. The DNS server resolves a small set of internal hostnames used during testing (e.g., core-1.data.com, server-1.data.com). The configured domain on infrastructure devices is data.com.

6.3 NTP

All infrastructure devices synchronize their clocks to Server-2 (10.10.30.20) over Network Time Protocol. NTP authentication is enabled with MD5 key 1 (key string "ccna") so that clients only accept time from a peer presenting the correct authentication key. Synchronized time is operationally important because syslog timestamps, AAA logs, and certificate validation all depend on consistent clocks across devices.

```
! On every internal infrastructure device
ntp authentication-key 1 md5 ccna
ntp trusted-key 1
ntp server 10.10.30.20 key 1
```

Verification commands: show ntp status (must report "Clock is synchronized") and show ntp associations (the configured server should be marked with an asterisk indicating it is the chosen master).

6.4 Syslog

All internal infrastructure devices are configured to forward log messages to the centralized Syslog collector on Server-1 (10.10.30.10) at debugging severity:

```
! On every internal infrastructure device
logging host 10.10.30.10
logging trap debugging
logging buffered 8192
```

Combined with synchronized clocks (NTP), this allows time-correlated analysis of events across the entire fabric from a single console. Verification is performed by triggering known events (interface flap via shutdown / no shutdown) and confirming that the corresponding messages arrive at the Syslog server with accurate timestamps.

7. Implementation

7.1 Device Inventory and Platform Models

Device	Role	Packet Tracer Model	Quantity
EDGE-RTR	WAN edge / NAT	Cisco ISR 2911 + 2× HWIC-1GE-SFP	1
CORE-1, CORE-2	Collapsed core (multilayer L3)	Cisco 3560-24PS	2
ACC-GF-1 to ACC-F5-1	Access layer	Cisco 2960-24TT	5 (reduced)
ISP-A, ISP-B	Simulated upstream providers	Cisco 2911 + HWIC-1GE-SFP	2
Server-1	DNS + Syslog + NTP source host	PT Server (generic)	1
Server-2	DHCP service + NTP master	PT Server (generic)	1
End hosts	PCs, IP phones, printers, laptops	PT generic endpoints	Multiple per floor
Access points	Wireless bridge for corporate/guest SSIDs	PT AP-PT	2 per floor (one per SSID)

7.2 Packet Tracer Reduction Strategy

The full deployment includes four access switches per floor for port-density reasons (twenty access switches total). The Packet Tracer model implements one representative access switch per floor (five total). This decision was made for two reasons: Packet Tracer becomes unstable with very large topologies, and the design verification objectives (HSRP failover, OSPF adjacency, DHCP relay, NAT) are fully demonstrated with one switch per floor. The additional fifteen switches are documented in the IP plan and port-assignment tables but not instantiated.

Specifically, in the reduced model:

- Each floor's representative access switch is configured as ACC-GF-1, ACC-F2-1, ACC-F3-1, ACC-F4-1, and ACC-F5-1. The configuration templates for the remaining switches (ACC-Fn-2 through ACC-Fn-4 on each floor) are identical except for the management SVI IP and the uplink port descriptions.
- The two cores are real devices. So is the edge router and both simulated ISPs.
- Server functions (DHCP, DNS, Syslog, NTP) are real — implemented on actual PT Server devices, with real scope, name resolution, and time synchronization verifiable in the simulation.

7.3 AP and Wireless Considerations

The Packet Tracer AP-PT device is a single-VLAN wireless bridge and does not support 802.1Q trunking or multiple SSIDs per radio. To deliver both a corporate SSID (WIFI_CORP, VLAN 60) and a guest SSID (WIFI_GUEST, VLAN 70) per floor, two AP-PT devices are deployed per floor: one access-port-connected into VLAN 60, the other into VLAN 70. In a production deployment

using a Cisco WLC and modern Aironet access points, a single AP would broadcast both SSIDs and the controller would tag each SSID's traffic into the appropriate VLAN at the controller. The two-AP arrangement is a simulator-specific compromise.

7.4 Build Order

Devices are configured in the following order to avoid management-plane lockout, ensure that dependent services are available when needed, and minimize disruptive convergence events:

1. Console access established on every device (physical or PT console).
2. Hostname, no ip domain-lookup, ip domain-name, and a local administrative user account on every device.
3. VLAN database created on every switch with vtp mode transparent and all VLAN IDs/names defined.
4. Trunks configured between access and cores and between cores: native VLAN 100, explicit allowed-VLAN lists, switchport nonegotiate.
5. Access ports configured: data and voice VLANs, PortFast on user-facing ports.
6. Spanning Tree explicit priorities applied on CORE-1 and CORE-2.
7. SVIs created on both cores with the planned IPs and HSRP groups; helper-address pointing to 10.10.30.20.
8. Routed interfaces (EDGE-CORE, EDGE-WAN, inter-core EtherChannel) configured.
9. Loopbacks configured for OSPF router-id and management.
10. OSPF process started with explicit router-id, passive-interface, and network statements.
11. Static default routes and NAT configured on EDGE-RTR; default-information originate enabled.
12. Server-1 and Server-2 NIC settings configured statically in the PT GUI.
13. DHCP scopes added on Server-2 via the PT Services panel.
14. NTP, Syslog, and DNS pointers added on every infrastructure device.
15. Hardening block applied last on every internal device: service password-encryption, enable secret, local user, VTY ACL, SSH v2, console hygiene.

8. Design Justification

This section consolidates the principal design decisions and prepares the team for likely panelist questions. Each subsection follows the pattern: Decision → Alternatives → Rationale → 30-second defense answer.

8.1 Collapsed Core vs. Full Three-Tier

A traditional three-tier hierarchy separates Core and Distribution into two physically distinct pairs of switches. For a single building with one distribution block, this physical separation adds cost and complexity without delivering measurable benefit. The collapsed core merges the Core and Distribution roles into one tier of multilayer switches, retaining all the logical functions (inter-VLAN routing, route summarization, first-hop redundancy) while reducing device count and operational overhead.

30-second answer: "Collapsed core is the industry-standard design for single-site enterprises of this scale. We retain all three logical layers conceptually — Access, Distribution, Core — but combine the Distribution and Core into one physical tier because we have one distribution block. The design still demonstrates HSRP, OSPF, EtherChannel, and STP root-bridge placement, and remains scalable by adding a true Core tier later if CAAP expands to multiple buildings."

8.2 HSRPv2 vs. VRRP vs. GLBP

VRRP is a vendor-neutral standard but offers no advantage over HSRPv2 in an all-Cisco environment. GLBP provides per-flow load-balancing across multiple gateways automatically but is Cisco-proprietary and introduces unnecessary operational complexity at this scale. HSRPv2 is mature, well-documented, supported across the entire Cisco line, and sufficient for the redundancy objectives of this project. Load distribution is achieved through explicit per-VLAN active/standby assignment rather than through GLBP's automatic mechanism, which makes the active path predictable for troubleshooting.

8.3 Odd/Even VLAN Load Split

Active responsibility for each VLAN is split by the parity of the VLAN identifier on the user VLANs (ODD on CORE-1, EVEN on CORE-2). Shared service VLANs (Servers, Management, Printers, both Wi-Fi VLANs) all follow CORE-1.

30-second answer: "Odd-numbered VLANs converge through CORE-1 and even-numbered VLANs through CORE-2 under normal operation, giving us automatic east-west load distribution without any dynamic protocol on top of HSRP. STP root-bridge placement is aligned to HSRP active so the Layer-2 path matches the Layer-3 gateway. Either core can serve any VLAN in a failure scenario because every SVI exists on both sides — only the priority differs."

8.4 OSPFv2 over EIGRP and Static Routing

Static routing at this scale is fragile: every link change would require manual updates on multiple devices, and any asymmetric path would be difficult to manage. EIGRP is Cisco-proprietary and although Packet Tracer supports it, OSPF is the more transferable choice for skill development and CCNA alignment. OSPFv2 is RFC 2328-standardized, vendor-neutral, and well-understood by anyone evaluating this design.

8.5 Single Edge Router with Floating Static Failover

A second edge router would have provided additional protection against router hardware failure, but at the cost of significantly more configuration (BGP or HSRP on the edge, route advertisement coordination, NAT statefulness). For an academic project where the failover behavior to be demonstrated is link/path failure rather than chassis failure, a single edge with two ISP uplinks and a floating static backup achieves the demonstration objective at a fraction of the complexity. The route preference is interface-state driven: when ISP-A's outgoing interface goes down, its static default route is removed and the floating static through ISP-B is installed.

8.6 Layer-3 EtherChannel Between Cores

The inter-core link is Layer-3 routed (no switchport, Port-channel 1 with /30 addressing), not a Layer-2 trunk. A Layer-2 trunk would have introduced inter-core paths into the spanning tree and tied inter-VLAN forwarding between the cores to STP convergence. A routed EtherChannel keeps the link in the routing domain, lets OSPF maintain a coherent topology, and uses LACP to bundle two members for both load distribution and fast member-failure recovery.

8.7 Native VLAN 100 (Discard) on Trunks

Default native VLAN 1 is the target of double-tag VLAN-hopping attacks. VLAN 100 has no SVI on either core, so untagged frames received on a trunk are dropped without routing.

8.8 VTP Transparent Mode

VTP transparent mode is chosen over server/client to eliminate the risk that a misconfigured client (with a higher revision number) erases the VLAN database campus-wide. The operational cost of defining VLANs per switch is negligible at this scale.

8.9 DHCP on a Centralized Server with Relay

DHCP runs centrally on Server-2, with ip helper-address configured on both cores (Active and Standby). Centralizing the service mirrors real enterprise practice and lets DHCP survive HSRP failover transparently.

8.10 Known Limitations

- Reduced device count in PT (one access switch per floor instead of four).
- No DHCP Snooping, Dynamic ARP Inspection, or IP Source Guard.
- No port-security in the simulated configuration (documented as a future enhancement).
- No 802.1X with RADIUS-based dynamic VLAN assignment.
- No QoS marking or queuing for voice traffic.
- Single edge router (single chassis is a single point of failure).
- Demonstration credentials only; no AAA server.
- IPv4 only; no IPv6 dual-stack.

8.11 Future Work

- Layer-2 hardening: DHCP Snooping, Dynamic ARP Inspection, IP Source Guard, and Port-Security with sticky MAC across all access ports.

- IEEE 802.1X with a RADIUS server for wired and wireless authentication and dynamic VLAN assignment.
- Out-of-band management network on a dedicated VLAN segregated from production.
- Border Gateway Protocol (BGP) with each ISP, replacing the static-default arrangement for true active-active internet ingress and ASN visibility.
- IPv6 dual-stack deployment across all VLANs.
- Zone-based firewall on the edge router for stateful inspection of inbound and outbound traffic.
- Quality of Service: DSCP marking at the access layer, priority queuing on uplinks, voice-specific shaping.
- Centralized monitoring: SNMP v3, NetFlow, and a NMS such as LibreNMS or Cisco DNA Center.
- A second edge router with HSRP between the two edges for chassis-level redundancy at the WAN.

9. Conclusion

This project designed and simulated a resilient multi-floor campus network for the New CAAP Building: collapsed-core architecture with HSRPv2 odd/even load split, OSPF Area 0, single edge router with dual-ISP failover, and centralized DHCP/DNS/NTP/Syslog services on two servers in VLAN 30.

Each principal design choice was evaluated against alternatives and justified in Section 9.

Known limitations and a future-work roadmap are in Section 9. The Packet Tracer simulation, configurations (Appendix C), IP/VLAN plan (Appendix A), and port map (Appendix B) together form the deliverable package.

References

- Cisco Systems. (2014). Campus LAN and Wireless LAN Design Guide. Cisco Validated Design.
- Moy, J. (1998). OSPF Version 2. RFC 2328. Internet Engineering Task Force.
- Li, T., Cole, B., Morton, P., & Li, D. (1998). Cisco Hot Standby Router Protocol (HSRP). RFC 2281. Internet Engineering Task Force.
- Nadas, S. (2010). Virtual Router Redundancy Protocol (VRRP) Version 3 for IPv4 and IPv6. RFC 5798. Internet Engineering Task Force.
- Arkko, J., Cotton, M., & Vegoda, L. (2010). IPv4 Address Blocks Reserved for Documentation. RFC 5737. Internet Engineering Task Force.
- Cisco Systems. (2020). Cisco IOS Security Configuration Guide. Cisco Press.
- Odom, W. (2020). CCNA 200-301 Official Cert Guide, Volumes 1 and 2. Cisco Press.
- Lammle, T. (2020). CCNA Certification Study Guide. Sybex.
- TIA-568.0-D (2015). Generic Telecommunications Cabling for Customer Premises. Telecommunications Industry Association.

Appendix A — IP and VLAN Plan

A.1 Master VLAN Table

VLAN	Name	Subnet	Mask	Gateway (VIP)	HSRP Active	DHCP?
11	DATA_GF	10.1.11.0	/24	10.1.11.1	CORE-1	Yes
12	DATA_F2	10.1.12.0	/24	10.1.12.1	CORE-2	Yes
13	DATA_F3	10.1.13.0	/24	10.1.13.1	CORE-1	Yes
14	DATA_F4	10.1.14.0	/24	10.1.14.1	CORE-2	Yes
15	DATA_F5	10.1.15.0	/24	10.1.15.1	CORE-1	Yes
21	VOICE_GF	10.2.21.0	/24	10.2.21.1	CORE-1	Yes
22	VOICE_F2	10.2.22.0	/24	10.2.22.1	CORE-2	Yes
23	VOICE_F3	10.2.23.0	/24	10.2.23.1	CORE-1	Yes
24	VOICE_F4	10.2.24.0	/24	10.2.24.1	CORE-2	Yes
25	VOICE_F5	10.2.25.0	/24	10.2.25.1	CORE-1	Yes
30	SERVERS	10.10.30.0	/26	10.10.30.1	CORE-1	No (static)
40	MGMT	10.10.40.0	/26	10.10.40.1	CORE-1	No (static)
50	PRINTERS	10.10.50.0	/27	10.10.50.1	CORE-1	Yes
60	WIFI_CORP	10.20.60.0	/23	10.20.60.1	CORE-1	Yes
70	WIFI_GUEST	10.20.70.0	/24	10.20.70.1	CORE-1	Yes (DNS 8.8.8.8)
100	NATIVE	(not routed)	—	(none)	—	No

A.2 Routed Transit and Loopback Subnets

Subnet	Purpose	Side A IP	Side B IP
10.99.0.0/30	EDGE-RTR ↔ CORE-1	EDGE G0/0: 10.99.0.1	CORE-1 F0/1: 10.99.0.2
10.99.0.4/30	EDGE-RTR ↔ CORE-2	EDGE G0/1: 10.99.0.5	CORE-2 F0/1: 10.99.0.6
10.99.0.8/30	CORE-1 ↔ CORE-2 (Port-channel 1)	CORE-1 Po1: 10.99.0.9	CORE-2 Po1: 10.99.0.10
10.99.0.51/32	EDGE-RTR Loopback0 / OSPF RID	—	—
10.99.0.52/32	CORE-1 Loopback0 / OSPF RID	—	—
10.99.0.53/32	CORE-2 Loopback0 / OSPF RID	—	—
203.0.113.0/30	ISP-A ↔ EDGE-RTR (WAN-1)	ISP-A G0/0/0: 203.0.113.1	EDGE G0/0/0: 203.0.113.2
203.0.113.4/30	ISP-B ↔ EDGE-RTR (WAN-2)	ISP-B G0/0/0: 203.0.113.5	EDGE G0/1/0: 203.0.113.6

203.0.113.200/29	NAT pool (outside global)	—	—
------------------	---------------------------	---	---

A.3 VLAN 30 (SERVERS) Address Map

IP	Device	Role
10.10.30.1	HSRP VIP (group 30)	Default gateway for VLAN 30 hosts
10.10.30.2	CORE-1 SVI Vlan30	Active for VLAN 30
10.10.30.3	CORE-2 SVI Vlan30	Standby for VLAN 30
10.10.30.10	Server-1	DNS + Syslog collector + NTP source host (static)
10.10.30.20	Server-2	DHCP service + NTP master (static)
10.10.30.11 – 10.10.30.62	(unused)	Reserved for future server expansion

A.4 VLAN 40 (MGMT) Address Map

IP	Device	Role
10.10.40.1	HSRP VIP (group 40)	Default gateway for MGMT VLAN
10.10.40.2	CORE-1 SVI Vlan40	Active for VLAN 40
10.10.40.3	CORE-2 SVI Vlan40	Standby for VLAN 40
10.10.40.21 – .24	ACC-GF-1 through ACC-GF-4	Mgmt SVIs on GF access switches
10.10.40.25 – .28	ACC-F2-1 through ACC-F2-4	Mgmt SVIs on F2 access switches
10.10.40.29 – .32	ACC-F3-1 through ACC-F3-4	Mgmt SVIs on F3 access switches
10.10.40.33 – .36	ACC-F4-1 through ACC-F4-4	Mgmt SVIs on F4 access switches
10.10.40.37 – .40	ACC-F5-1 through ACC-F5-4	Mgmt SVIs on F5 access switches

A.5 DHCP Scope Reference (Server-2 GUI)

Scopes are configured via the Packet Tracer Services panel on Server-2. Each scope below corresponds to one VLAN; the Default Gateway value is the HSRP VIP for that VLAN, ensuring that hosts receive a gateway that survives HSRP failover.

Pool Name	Network	Mask	Default Gateway	DNS	Start IP	Options
DATA_GF	10.1.11.0	255.255.255.0	10.1.11.1	10.10.30.10	10.1.11.11	—
DATA_F2	10.1.12.0	255.255.255.0	10.1.12.1	10.10.30.10	10.1.12.11	—
DATA_F3	10.1.13.0	255.255.255.0	10.1.13.1	10.10.30.10	10.1.13.11	—
DATA_F4	10.1.14.0	255.255.255.0	10.1.14.1	10.10.30.10	10.1.14.11	—
DATA_F5	10.1.15.0	255.255.255.0	10.1.15.1	10.10.30.10	10.1.15.11	—
VOICE_GF	10.2.21.0	255.255.255.0	10.2.21.1	10.10.30.10	10.2.21.11	TFTP 10.10.30.10
VOICE_F2	10.2.22.0	255.255.255.0	10.2.22.1	10.10.30.10	10.2.22.11	TFTP 10.10.30.10
VOICE_F3	10.2.23.0	255.255.255.0	10.2.23.1	10.10.30.10	10.2.23.11	TFTP 10.10.30.10

VOICE_F4	10.2.24.0	255.255.255.0	10.2.24.1	10.10.30.10	10.2.24.11	TFTP 10.10.30.10
VOICE_F5	10.2.25.0	255.255.255.0	10.2.25.1	10.10.30.10	10.2.25.11	TFTP 10.10.30.10
PRINTERS	10.10.50.0	255.255.255.224	10.10.50.1	10.10.30.10	10.10.50.11	—
WIFI_CORP	10.20.60.0	255.255.254.0	10.20.60.1	10.10.30.10	10.20.60.11	—
WIFI_GUEST	10.20.70.0	255.255.255.0	10.20.70.1	8.8.8.8	10.20.70.11	Public DNS

VLAN 30 (SERVERS) and VLAN 40 (MGMT) are not in DHCP; all addresses in those VLANs are statically configured.

Appendix B — Port-by-Port Connection Map

B.1 WAN and Routed Transit

From	Interface	To	Interface	Subnet	Media
EDGE-RTR	G0/0/0	ISP-A	G0/0/0	203.0.113.0/30	SMF + GLC-LH-SMD
EDGE-RTR	G0/1/0	ISP-B	G0/0/0	203.0.113.4/30	SMF + GLC-LH-SMD
EDGE-RTR	G0/0	CORE-1	F0/1	10.99.0.0/30	Cat6 STC
EDGE-RTR	G0/1	CORE-2	F0/1	10.99.0.4/30	Cat6 STC
CORE-1	G0/1 (Po1)	CORE-2	G0/1 (Po1)	10.99.0.8/30	Cat6 STC or fiber
CORE-1	G0/2 (Po1)	CORE-2	G0/2 (Po1)	(same Po1)	Cat6 STC or fiber

B.2 Access-Switch Uplinks to Cores (Reduced Model)

Access Switch	G0/1 to CORE-1	G0/2 to CORE-2	Native VLAN	Allowed VLANs
ACC-GF-1	F0/2	F0/2	100	11, 21, 30, 40, 50, 60, 70, 100
ACC-F2-1	F0/6	F0/6	100	12, 22, 40, 50, 60, 70, 100
ACC-F3-1	F0/10	F0/10	100	13, 23, 40, 50, 60, 70, 100
ACC-F4-1	F0/14	F0/14	100	14, 24, 40, 50, 60, 70, 100
ACC-F5-1	F0/18	F0/18	100	15, 25, 40, 50, 60, 70, 100

In the full design, each floor has four access switches (ACC-Fn-1 through ACC-Fn-4), each occupying one consecutive core port pair (e.g., GF uses cores F0/2 through F0/5; F2 uses F0/6 through F0/9; F3 uses F0/10 through F0/13; F4 uses F0/14 through F0/17; F5 uses F0/18 through F0/21). Core ports F0/22 through F0/24 are administratively shut.

B.3 Access-Switch Local Port Allocation

Switch	Server Ports	Printer Ports	AP Ports	User Ports (Data+Voice)
ACC-GF-1	F0/1 – F0/2 (VLAN 30)	F0/3 – F0/7 (VLAN 50)	F0/8 = VLAN 60 (CORP); F0/9 = VLAN 70 (GUEST)	F0/10 – F0/24 (VLAN 11 + voice VLAN 21)
ACC-F2-1	—	F0/1 – F0/3 (VLAN 50)	F0/4 = VLAN 60; F0/5 = VLAN 70	F0/6 – F0/24 (VLAN 12 + voice VLAN 22)
ACC-F3-1	—	F0/1 – F0/3 (VLAN 50)	F0/4 = VLAN 60; F0/5 = VLAN 70	F0/6 – F0/24 (VLAN 13 + voice VLAN 23)
ACC-F4-1	—	F0/1 – F0/3 (VLAN 50)	F0/4 = VLAN 60; F0/5 = VLAN 70	F0/6 – F0/24 (VLAN 14 + voice VLAN 24)
ACC-F5-1	—	F0/1 – F0/3 (VLAN 50)	F0/4 = VLAN 60; F0/5 = VLAN 70	F0/6 – F0/24 (VLAN 15 + voice VLAN 25)

B.4 Server Connectivity

Server	NIC IP	Mask	Gateway	DNS	Switch Port	VLAN
Server-1	10.10.30.10	255.255.255.192	10.10.30.1	10.10.30.10 (self)	ACC-GF-1 F0/1	30
Server-2	10.10.30.20	255.255.255.192	10.10.30.1	10.10.30.10	ACC-GF-1 F0/2	30

Appendix C — Device Configurations

The complete running configurations for every internal device are reproduced below. These configurations are the authoritative source — any apparent discrepancy between the body of this report and these configurations should be resolved in favor of the configuration text.

C.1 EDGE-RTR

```
! =====
! EDGE-RTR - WAN edge router (FINAL)
! Role: NAT (pool overload), OSPF Area 0 to cores,
!       dual static defaults to ISPs (interface-state failover)
! =====

enable
configure terminal

hostname EDGE-RTR
no ip domain-lookup
ip domain name data.com

! =====
! ===== Step 1: Interfaces =====
! =====

interface loopback 0
  description Router-id and management
  ip address 10.99.0.51 255.255.255.255
exit

interface g0/0
  description Link to CORE-1 F0/1
  ip address 10.99.0.1 255.255.255.252
  ip ospf network point-to-point
  ip nat inside
  no shutdown
exit

interface g0/1
  description Link to CORE-2 F0/1
  ip address 10.99.0.5 255.255.255.252
  ip ospf network point-to-point
  ip nat inside
  no shutdown
exit

interface g0/0/0
  description Link to ISP-A G0/0/0 (primary)
  ip address 203.0.113.2 255.255.255.252
  ip nat outside
  no shutdown
exit
```

```

interface g0/1/0
  description Link to ISP-B G0/0/0 (secondary)
  ip address 203.0.113.6 255.255.255.252
  ip nat outside
  no shutdown
exit

! =====
! ===== Step 2: OSPF =====
! =====
router ospf 1
  router-id 10.99.0.51
  passive-interface loopback 0
  network 10.99.0.51 0.0.0.0 area 0
  network 10.99.0.1 0.0.0.0 area 0
  network 10.99.0.5 0.0.0.0 area 0
  default-information originate
exit

! =====
! ===== Step 3: Default routes (primary + floating backup) =====
! =====
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip route 0.0.0.0 0.0.0.0 203.0.113.5 5

! =====
! ===== Step 4: NAT (pool overload) =====
! =====
access-list 2 permit 10.1.0.0 0.0.255.255
access-list 2 permit 10.2.0.0 0.0.255.255
access-list 2 permit 10.10.0.0 0.0.255.255
access-list 2 permit 10.20.0.0 0.0.255.255

ip nat pool POOL1 203.0.113.200 203.0.113.207 netmask 255.255.255.248
ip nat inside source list 2 pool POOL1 overload

! =====
! ===== Step 5: Logging, NTP, DNS =====
! =====
ip name-server 10.10.30.10

logging 10.10.30.10
logging trap debugging
logging buffered 8192

ntp authentication-key 1 md5 ccna
ntp trusted-key 1
ntp server 10.10.30.20 key 1

no cdp run
lldp run

! =====

```

```
! ===== Step 6: Auth, SSH, VTY ACL (final hardening) =====
! =====
service password-encryption
enable secret password
username password privilege 15 secret password

access-list 1 permit 10.10.40.0 0.0.0.63
access-list 1 deny any

crypto key generate rsa
512
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3

line console 0
 login local
 logging synchronous
 exec-timeout 10 0
exit

line vty 0 4
 login local
 transport input ssh
 access-class 1 in
 exec-timeout 10 0
exit

end
write memory
```

C.2 CORE-1

```
! =====
! CORE-1 – Collapsed Core L3 switch (FINAL)
! Role: SVIs for all VLANs, HSRP active for ODD VLANs and shared,
!       OSPF Area 0, STP root for ODD VLANs and shared
! =====

enable
configure terminal

hostname CORE-1
no ip domain-lookup
ip domain name data.com

! =====
! ===== Step 1: Enable IP routing =====
! =====
ip routing

! =====
! ===== Step 2: VTP and VLANs =====
! =====
vtp version 2
vtp mode transparent
vtp domain-name data.com

vlan 11
    name DATA_GF
vlan 12
    name DATA_F2
vlan 13
    name DATA_F3
vlan 14
    name DATA_F4
vlan 15
    name DATA_F5
vlan 21
    name VOICE_GF
vlan 22
    name VOICE_F2
vlan 23
    name VOICE_F3
vlan 24
    name VOICE_F4
vlan 25
    name VOICE_F5
vlan 30
    name SERVERS
vlan 40
    name MGMT
vlan 50
    name PRINTERS
```

```

vlan 60
  name WIFI_CORP
vlan 70
  name WIFI_GUEST
vlan 100
  name NATIVE
exit

! =====
! ===== Step 3: Interfaces =====
! =====

interface loopback 0
  description Router-id
  ip address 10.99.0.52 255.255.255.255
exit

interface f0/1
  description Link to EDGE-RTR G0/0
  no switchport
  ip address 10.99.0.2 255.255.255.252
  ip ospf network point-to-point
  no shutdown
exit

interface range g0/1 - 2
  description Routed link to CORE-2 (Po1 members)
  no switchport
  channel-group 1 mode active
  no shutdown
exit

interface port-channel 1
  description Routed L3 EtherChannel to CORE-2
  no switchport
  ip address 10.99.0.9 255.255.255.252
  ip ospf network point-to-point
exit

interface range f0/2 - 5
  description Trunk to ACC-GF-1..4 (their G0/1)
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport nonegotiate
  switchport trunk native vlan 100
  switchport trunk allowed vlan 11,21,30,40,50,60,70,100
  no shutdown
exit

interface range f0/6 - 9
  description Trunk to ACC-F2-1..4 (their G0/1)
  switchport trunk encapsulation dot1q
  switchport mode trunk

```

```

switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 12,22,40,50,60,70,100
no shutdown
exit

interface range f0/10 - 13
description Trunk to ACC-F3-1..4 (their G0/1)
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 13,23,40,50,60,70,100
no shutdown
exit

interface range f0/14 - 17
description Trunk to ACC-F4-1..4 (their G0/1)
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 14,24,40,50,60,70,100
no shutdown
exit

interface range f0/18 - 21
description Trunk to ACC-F5-1..4 (their G0/1)
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 15,25,40,50,60,70,100
no shutdown
exit

interface range f0/22 - 24
description Unused – shutdown
shutdown
exit

! =====
! ===== Step 4: SVIs with HSRP =====
! =====
! CORE-1 ACTIVE for: ODD VLANs (11,13,15,21,23,25) and shared (30,40,50,60,70)
! CORE-1 ST

ANDBY for: EVEN VLANs (12,14,22,24)

interface vlan 11
description DATA_GF – Core-1 active (ODD)
ip address 10.1.11.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2

```

```
standby 11 ip 10.1.11.1
standby 11 priority 110
standby 11 preempt
no shutdown
exit

interface vlan 12
description DATA_F2 – Core-1 standby (EVEN)
ip address 10.1.12.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 12 ip 10.1.12.1
no shutdown
exit

interface vlan 13
description DATA_F3 – Core-1 active (ODD)
ip address 10.1.13.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 13 ip 10.1.13.1
standby 13 priority 110
standby 13 preempt
no shutdown
exit

interface vlan 14
description DATA_F4 – Core-1 standby (EVEN)
ip address 10.1.14.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 14 ip 10.1.14.1
no shutdown
exit

interface vlan 15
description DATA_F5 – Core-1 active (ODD)
ip address 10.1.15.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 15 ip 10.1.15.1
standby 15 priority 110
standby 15 preempt
no shutdown
exit

interface vlan 21
description VOICE_GF – Core-1 active (ODD)
ip address 10.2.21.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 21 ip 10.2.21.1
standby 21 priority 110
```



```
standby 21 preempt
no shutdown
exit

interface vlan 22
description VOICE_F2 – Core-1 standby (EVEN)
ip address 10.2.22.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 22 ip 10.2.22.1
no shutdown
exit

interface vlan 23
description VOICE_F3 – Core-1 active (ODD)
ip address 10.2.23.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 23 ip 10.2.23.1
standby 23 priority 110
standby 23 preempt
no shutdown
exit

interface vlan 24
description VOICE_F4 – Core-1 standby (EVEN)
ip address 10.2.24.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 24 ip 10.2.24.1
no shutdown
exit

interface vlan 25
description VOICE_F5 – Core-1 active (ODD)
ip address 10.2.25.2 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 25 ip 10.2.25.1
standby 25 priority 110
standby 25 preempt
no shutdown
exit

interface vlan 30
description SERVERS – Core-1 active
ip address 10.10.30.2 255.255.255.192
standby version 2
standby 30 ip 10.10.30.1
standby 30 priority 110
standby 30 preempt
no shutdown
exit
```

```

interface vlan 40
  description MGMT – Core-1 active
  ip address 10.10.40.2 255.255.255.192
  standby version 2
  standby 40 ip 10.10.40.1
  standby 40 priority 110
  standby 40 preempt
  no shutdown
exit

interface vlan 50
  description PRINTERS – Core-1 active
  ip address 10.10.50.2 255.255.255.224
  ip helper-address 10.10.30.20
  standby version 2
  standby 50 ip 10.10.50.1
  standby 50 priority 110
  standby 50 preempt
  no shutdown
exit

interface vlan 60
  description WIFI_CORP – Core-1 active
  ip address 10.20.60.2 255.255.254.0
  ip helper-address 10.10.30.20
  standby version 2
  standby 60 ip 10.20.60.1
  standby 60 priority 110
  standby 60 preempt
  no shutdown
exit

interface vlan 70
  description WIFI_GUEST – Core-1 active
  ip address 10.20.70.2 255.255.255.0
  ip helper-address 10.10.30.20
  standby version 2
  standby 70 ip 10.20.70.1
  standby 70 priority 110
  standby 70 preempt
  no shutdown
exit

! =====
! ===== Step 5: OSPF =====
! =====

router ospf

f 1
  router-id 10.99.0.52
  passive-interface loopback 0
  passive-interface vlan 11
  passive-interface vlan 12

```

```

passive-interface vlan 13
passive-interface vlan 14
passive-interface vlan 15
passive-interface vlan 21
passive-interface vlan 22
passive-interface vlan 23
passive-interface vlan 24
passive-interface vlan 25
passive-interface vlan 30
passive-interface vlan 40
passive-interface vlan 50
passive-interface vlan 60
passive-interface vlan 70
network 10.99.0.52 0.0.0.0 area 0
network 10.99.0.2 0.0.0.0 area 0
network 10.99.0.9 0.0.0.0 area 0
network 10.1.11.2 0.0.0.0 area 0
network 10.1.12.2 0.0.0.0 area 0
network 10.1.13.2 0.0.0.0 area 0
network 10.1.14.2 0.0.0.0 area 0
network 10.1.15.2 0.0.0.0 area 0
network 10.2.21.2 0.0.0.0 area 0
network 10.2.22.2 0.0.0.0 area 0
network 10.2.23.2 0.0.0.0 area 0
network 10.2.24.2 0.0.0.0 area 0
network 10.2.25.2 0.0.0.0 area 0
network 10.10.30.2 0.0.0.0 area 0
network 10.10.40.2 0.0.0.0 area 0
network 10.10.50.2 0.0.0.0 area 0
network 10.20.60.2 0.0.0.0 area 0
network 10.20.70.2 0.0.0.0 area 0
exit

! =====
! ===== Step 6: Spanning Tree =====
! =====
spanning-tree mode rapid-pvst
spanning-tree vlan 11,13,15,21,23,25,30,40,50,60,70,100 priority 0
spanning-tree vlan 12,14,22,24 priority 4096

! =====
! ===== Step 7: Logging, NTP, DNS =====
! =====
ip name-server 10.10.30.10

logging 10.10.30.10
logging trap debugging
logging buffered 8192

ntp authentication-key 1 md5 ccna
ntp trusted-key 1
ntp server 10.10.30.20 key 1

```

```
no cdp run
lldp run

! =====
! ===== Step 8: Auth, SSH, VTY ACL (final hardening) =====
! =====
service password-encryption
enable secret password
username password privilege 15 secret password

! Restrict SSH to MGMT VLAN sources only
access-list 1 permit 10.10.40.0 0.0.0.63
access-list 1 deny any

! SSH server
crypto key generate rsa
512
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3

line console 0
login local
logging synchronous
exec-timeout 10 0
exit

line vty 0 15
login local
transport input ssh
access-class 1 in
exec-timeout 10 0
exit

end
write memory
```

C.3 CORE-2

```
! =====
! CORE-2 – Collapsed Core L3 switch (FINAL, mirror of CORE-1)
! Role: HSRP active for EVEN VLANs (12,14,22,24),
!       HSRP standby for ODD VLANs and shared
! =====

enable
configure terminal

hostname CORE-2
no ip domain-lookup
ip domain name data.com

! =====
! ===== Step 1: Enable IP routing =====
! =====
ip routing

! =====
! ===== Step 2: VTP and VLANs =====
! =====
vtp version 2
vtp mode transparent
vtp domain-name data.com

vlan 11
    name DATA_GF
vlan 12
    name DATA_F2
vlan 13
    name DATA_F3
vlan 14
    name DATA_F4
vlan 15
    name DATA_F5
vlan 21
    name VOICE_GF
vlan 22
    name VOICE_F2
vlan 23
    name VOICE_F3
vlan 24
    name VOICE_F4
vlan 25
    name VOICE_F5
vlan 30
    name SERVERS
vlan 40
    name MGMT
vlan 50
    name PRINTERS
```

```

vlan 60
  name WIFI_CORP
vlan 70
  name WIFI_GUEST
vlan 100
  name NATIVE
exit

! =====
! ===== Step 3: Interfaces =====
! =====

interface loopback 0
  description Router-id
  ip address 10.99.0.53 255.255.255.255
exit

interface f0/1
  description Link to EDGE-RTR G0/1
  no switchport
  ip address 10.99.0.6 255.255.255.252
  ip ospf network point-to-point
  no shutdown
exit

interface range g0/1 - 2
  description Routed link to CORE-1 (Po1 members)
  no switchport
  channel-group 1 mode active
  no shutdown
exit

interface port-channel 1
  description Routed L3 EtherChannel to CORE-1
  no switchport
  ip address 10.99.0.10 255.255.255.252
  ip ospf network point-to-point
exit

interface range f0/2 - 5
  description Trunk to ACC-GF-1..4 (their G0/2)
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport nonegotiate
  switchport trunk native vlan 100
  switchport trunk allowed vlan 11,21,30,40,50,60,70,100
  no shutdown
exit

interface range f0/6 - 9
  description Trunk to ACC-F2-1..4 (their G0/2)
  switchport trunk encapsulation dot1q
  switchport mode trunk

```

```
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 12,22,40,50,60,70,100
no shutdown
exit
```

```
interface range f0/10 - 13
description Trunk to ACC-F3-1..4 (their G0/2)
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 13,23,40,50,60,70,100
no shutdown
exit
```

```
interface range f0/14 - 17
description Trunk to ACC-F4-1..4 (their G0/2)
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 14,24,40,50,60,70,100
no shutdown
exit
```

```
interface range f0/18 - 21
description Trunk to ACC-F5-1..4 (their G0/2)
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 15,25,40,50,60,70,100
no shutdown
exit
```

```
interface range f0/22 - 24
description Unused – shutdown
shutdown
exit
```

```
! =====
! ===== Step 4: SVIs with HSRP (mirror of CORE-1) =====
! =====
```

```
interface vlan 11
description DATA_GF – Core-2 standby (ODD)
ip address 10
.1.11.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 11 ip 10.1.11.1
no shutdown
```

```
exit

interface vlan 12
description DATA_F2 – Core-2 active (EVEN)
ip address 10.1.12.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 12 ip 10.1.12.1
standby 12 priority 110
standby 12 preempt
no shutdown
exit

interface vlan 13
description DATA_F3 – Core-2 standby (ODD)
ip address 10.1.13.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 13 ip 10.1.13.1
no shutdown
exit

interface vlan 14
description DATA_F4 – Core-2 active (EVEN)
ip address 10.1.14.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 14 ip 10.1.14.1
standby 14 priority 110
standby 14 preempt
no shutdown
exit

interface vlan 15
description DATA_F5 – Core-2 standby (ODD)
ip address 10.1.15.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 15 ip 10.1.15.1
no shutdown
exit

interface vlan 21
description VOICE_GF – Core-2 standby (ODD)
ip address 10.2.21.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 21 ip 10.2.21.1
no shutdown
exit

interface vlan 22
description VOICE_F2 – Core-2 active (EVEN)
```



```
ip address 10.2.22.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 22 ip 10.2.22.1
standby 22 priority 110
standby 22 preempt
no shutdown
exit

interface vlan 23
description VOICE_F3 – Core-2 standby (ODD)
ip address 10.2.23.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 23 ip 10.2.23.1
no shutdown
exit

interface vlan 24
description VOICE_F4 – Core-2 active (EVEN)
ip address 10.2.24.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 24 ip 10.2.24.1
standby 24 priority 110
standby 24 preempt
no shutdown
exit

interface vlan 25
description VOICE_F5 – Core-2 standby (ODD)
ip address 10.2.25.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 25 ip 10.2.25.1
no shutdown
exit

interface vlan 30
description SERVERS – Core-2 standby
ip address 10.10.30.3 255.255.255.192
standby version 2
standby 30 ip 10.10.30.1
no shutdown
exit

interface vlan 40
description MGMT – Core-2 standby
ip address 10.10.40.3 255.255.255.192
standby version 2
standby 40 ip 10.10.40.1
no shutdown
exit
```

```

interface vlan 50
description PRINTERS – Core-2 standby
ip address 10.10.50.3 255.255.255.224
ip helper-address 10.10.30.20
standby version 2
standby 50 ip 10.10.50.1
no shutdown
exit

interface vlan 60
description WIFI_CORP – Core-2 standby
ip address 10.20.60.3 255.255.254.0
ip helper-address 10.10.30.20
standby version 2
standby 60 ip 10.20.60.1
no shutdown
exit

interface vlan 70
description WIFI_GUEST – Core-2 standby
ip address 10.20.70.3 255.255.255.0
ip helper-address 10.10.30.20
standby version 2
standby 70 ip 10.20.70.1
no shutdown
exit

! =====
! ===== Step 5: OSPF =====
! =====

router ospf 1
router-id 10.99.0.53
passive-interface loopback 0
passive-interface vlan 11
passive-interface vlan 12
passive-interface vlan 13
passive-interface vlan 14
passive-interface vlan 15
passive-interface vlan 21
passive-interface vlan 22
passive-interface vlan 23
passive-interface vlan 24
passive-interface vlan 25
passive-interface vlan 30
passive-interface vlan 40
passive-interface vlan 50
passive-inter

face vlan 60
passive-interface vlan 70
network 10.99.0.53 0.0.0.0 area 0
network 10.99.0.6 0.0.0.0 area 0
network 10.99.0.10 0.0.0.0 area 0

```

```

network 10.1.11.3 0.0.0.0 area 0
network 10.1.12.3 0.0.0.0 area 0
network 10.1.13.3 0.0.0.0 area 0
network 10.1.14.3 0.0.0.0 area 0
network 10.1.15.3 0.0.0.0 area 0
network 10.2.21.3 0.0.0.0 area 0
network 10.2.22.3 0.0.0.0 area 0
network 10.2.23.3 0.0.0.0 area 0
network 10.2.24.3 0.0.0.0 area 0
network 10.2.25.3 0.0.0.0 area 0
network 10.10.30.3 0.0.0.0 area 0
network 10.10.40.3 0.0.0.0 area 0
network 10.10.50.3 0.0.0.0 area 0
network 10.20.60.3 0.0.0.0 area 0
network 10.20.70.3 0.0.0.0 area 0
exit

! =====
! ===== Step 6: Spanning Tree =====
! =====
spanning-tree mode rapid-pvst
spanning-tree vlan 12,14,22,24 priority 0
spanning-tree vlan 11,13,15,21,23,25,30,40,50,60,70,100 priority 4096

! =====
! ===== Step 7: Logging, NTP, DNS =====
! =====
ip name-server 10.10.30.10

logging 10.10.30.10
logging trap debugging
logging buffered 8192

ntp authentication-key 1 md5 ccna
ntp trusted-key 1
ntp server 10.10.30.20 key 1

no cdp run
lldp run

! =====
! ===== Step 8: Auth, SSH, VTY ACL (final hardening) =====
! =====
service password-encryption
enable secret password
username password privilege 15 secret password

access-list 1 permit 10.10.40.0 0.0.0.63
access-list 1 deny any

crypto key generate rsa
512
ip ssh version 2

```

```
ip ssh time-out 60
ip ssh authentication-retries 3

line console 0
  login local
  logging synchronous
  exec-timeout 10 0
exit

line vty 0 15
  login local
  transport input ssh
  access-class 1 in
  exec-timeout 10 0
exit

end
write memory
```

C.4 ACC-GF-1 (template for GF access switches)

```
! =====
! ACC-GF-1 – Ground Floor access switch (FINAL)
! Role: L2 access – servers (this switch only), printers, AP, users
! =====
! Port layout (24 ports):
!   F0/1-2   – Servers in VLAN 30
!   F0/3-7   – Printers in VLAN 50
!   F0/8     – AP-CORP (VLAN 60)
!   F0/9-24  – User access (DATA_GF + VOICE_GF)
! =====

enable
configure terminal

hostname ACC-GF-1
no ip domain-lookup
ip domain name data.com

! ===== Step 1: VTP and VLANs =====
vtp version 2
vtp mode transparent
vtp domain-name data.com

vlan 11
  name DATA_GF
vlan 21
  name VOICE_GF
vlan 30
  name SERVERS
vlan 40
  name MGMT
vlan 50
  name PRINTERS
vlan 60
  name WIFI_CORP
vlan 70
  name WIFI_GUEST
vlan 100
  name NATIVE
exit

! ===== Step 2: Spanning Tree =====
spanning-tree mode rapid-pvst

! ===== Step 3: Uplinks =====
interface g0/1
  description Uplink to CORE-1 F0/2
  switchport mode trunk
  switchport nonegotiate
  switchport trunk native vlan 100
  switchport trunk allowed vlan 11,21,30,40,50,60,70,100
```

```

no shutdown
exit

interface g0/2
description Uplink to CORE-2 F0/2
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 100
switchport trunk allowed vlan 11,21,30,40,50,60,70,100
no shutdown
exit

! ===== Step 4: Server ports (F0/1-2) =====
interface range f0/1 - 2
description Servers in VLAN 30
switchport mode access
switchport access vlan 30
switchport nonegotiate
spanning-tree portfast
spanning-tree bpduguard enable
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security maximum 2
no shutdown
exit

! ===== Step 5: Printer ports (F0/3-7) =====
interface range f0/3 - 7
description Network printers in VLAN 50
switchport mode access
switchport access vlan 50
switchport nonegotiate
spanning-tree portfast
spanning-tree bpduguard enable
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
no shutdown
exit

! ===== Step 6: AP ports =====
! NOTE: The PT AP-PT cannot trunk; one AP per SSID per floor.
! F0/8 = AP-CORP (VLAN 60), F0/9 reassigned to AP-GUEST (VLAN 70).
! No port-security on AP ports – wireless clients all share the AP MAC,
! and PortFast/BPDU-guard would falsely trip in some PT versions.
interface f0/8
description AP-CORP – WIFI_CORP (VLAN 60)
switchport mode access
switchport access vlan 60
switchport nonegotiate
spanning-tree portfast
no shutdown

```

```

exit

interface f0/9
description AP-GUEST - WIFI_GUEST (VLAN 70)
switchport mode access
switchport access vlan 70
switchport nonegotiate
spanning-tree portfast
no shutdown
exit

! ===== Step 7: User access ports (F0/10-24) =====
interface range f0/10 - 24
description User access - DATA_GF + VOICE_GF
switchport mode access
switchport nonegotiate
switchport access vlan 11
switchport voice vlan 21
spanning-tree portfast
spanning-tree bpduguard enable
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security maximum 3
no shutdown
exit

! ===== Step 8: Errdisable auto-recovery =====
errdisable recovery cause bpduguard
errdisable recovery cause psecure-violation
errdisable recovery interval 300

! ===== Step 9: Management SVI (VLAN 40) =====
! IP plan: ACC-GF-1 = 10.10.40.21
interface
vlan 40
description Management interface
ip address 10.10.40.21 255.255.255.192
no shutdown
exit

ip default-gateway 10.10.40.1

! ===== Step 10: Logging, NTP, DNS =====
ip name-server 10.10.30.10

logging 10.10.30.10
logging trap debugging
logging buffered 8192

ntp authentication-key 1 md5 ccna
ntp trusted-key 1
ntp server 10.10.30.20 key 1

```

```

no cdp run
lldp run

! =====
! ===== Step 11: Auth, SSH, VTY ACL (final hardening) =====
! =====
service password-encryption
enable secret password
username password privilege 15 secret password

access-list 1 permit 10.10.40.0 0.0.0.63
access-list 1 deny any

crypto key generate rsa
512
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3

line console 0
login local
logging synchronous
exec-timeout 10 0
exit

line vty 0 15
login local
transport input ssh
access-class 1 in
exec-timeout 10 0
exit

end
write memory

! =====
! NOTES FOR ACC-GF-2, ACC-GF-3, ACC-GF-4
! =====
! Identical EXCEPT:
!   hostname: ACC-GF-2 / ACC-GF-3 / ACC-GF-4
!   VLAN 40 IP: 10.10.40.22 / .23 / .24
!   Uplink port descriptions: ACC-GF-2 → CORE-1/2 F0/3,
!                             ACC-GF-3 → F0/4,
!                             ACC-GF-4 → F0/5
!   ACC-GF-2..4 do NOT have server ports.
!   Convert F0/1-2 to user access ports (same config as F0/10-24).
! =====

```


C.5 ISP-A

```
! =====
! ISP-A – Simulated upstream provider (primary, FINAL)
! =====
! NOTE: ISPs are simulation stubs outside the enterprise admin domain.
! No internal auth/SSH hardening – only baseline console hygiene.
! =====

enable
configure terminal

hostname ISP-A
no ip domain-lookup

interface g0/0/0
  description Link to EDGE-RTR G0/0/0
  ip address 203.0.113.1 255.255.255.252
  no shutdown
exit

interface loopback 0
  description Simulated internet host (ISP-A path)
  ip address 8.8.8.8 255.255.255.255
exit

! Return route for the NAT pool (so ping replies can come back)
ip route 203.0.113.200 255.255.255.248 203.0.113.2

line console 0
  logging synchronous
  exec-timeout 30 0
exit

end
write memory
```

C.6 ISP-B

```
! =====
! ISP-B – Simulated upstream provider (secondary / failover, FINAL)
! =====

enable
configure terminal

hostname ISP-B
no ip domain-lookup

interface g0/0/0
  description Link to EDGE-RTR G0/1/0
  ip address 203.0.113.5 255.255.255.252
  no shutdown
exit

interface loopback 0
  description Simulated internet host (ISP-B path)
  ip address 1.1.1.1 255.255.255.255
exit

! Return route for the NAT pool
ip route 203.0.113.200 255.255.255.248 203.0.113.6

line console 0
  logging synchronous
  exec-timeout 30 0
exit

end
write memory
```

C.7 DHCP Server Scopes (Server-2 GUI)

```
! =====
! DHCP-SERVER (Server-2 in VLAN 30) – Scope Reference
! =====
! Configure scopes via Server-2 GUI:
!   Services > DHCP > add each pool below > Save
!
! VLAN 30 (10.10.30.0/26) plan – usable .1 through .62:
!   .1   Gateway (HSRP VIP, owned by CORE-1 normally)
!   .2   CORE-1 SVI
!   .3   CORE-2 SVI
!   .10  Server-1 (DNS + SYSLOG + NTP source) – static on server NIC
!   .20  Server-2 (DHCP) – static on server NIC, gateway = 10.10.30.1
! =====

=====
SCOPE: DATA_GF (VLAN 11)
  Pool Name:      DATA_GF
  Default Gateway: 10.1.11.1
  DNS Server:     10.10.30.10
  Start IP:       10.1.11.11
  Subnet Mask:    255.255.255.0
  Maximum Users:  240

SCOPE: DATA_F2 (VLAN 12)
  Pool Name:      DATA_F2
  Default Gateway: 10.1.12.1
  DNS Server:     10.10.30.10
  Start IP:       10.1.12.11
  Subnet Mask:    255.255.255.0
  Maximum Users:  240

SCOPE: DATA_F3 (VLAN 13)
  Pool Name:      DATA_F3
  Default Gateway: 10.1.13.1
  DNS Server:     10.10.30.10
  Start IP:       10.1.13.11
  Subnet Mask:    255.255.255.0
  Maximum Users:  240

SCOPE: DATA_F4 (VLAN 14)
  Pool Name:      DATA_F4
  Default Gateway: 10.1.14.1
  DNS Server:     10.10.30.10
  Start IP:       10.1.14.11
  Subnet Mask:    255.255.255.0
  Maximum Users:  240

SCOPE: DATA_F5 (VLAN 15)
  Pool Name:      DATA_F5
  Default Gateway: 10.1.15.1
  DNS Server:     10.10.30.10
```

Start IP: 10.1.15.11
Subnet Mask: 255.255.255.0
Maximum Users: 240

SCOPE: VOICE_GF (VLAN 21)

Pool Name: VOICE_GF
Default Gateway: 10.2.21.1
DNS Server: 10.10.30.10
Start IP: 10.2.21.11
Subnet Mask: 255.255.255.0
Maximum Users: 240
TFTP Server: 10.10.30.10

SCOPE: VOICE_F2 (VLAN 22)

Pool Name: VOICE_F2
Default Gateway: 10.2.22.1
DNS Server: 10.10.30.10
Start IP: 10.2.22.11
Subnet Mask: 255.255.255.0
Maximum Users: 240
TFTP Server: 10.10.30.10

SCOPE: VOICE_F3 (VLAN 23)

Pool Name: VOICE_F3
Default Gateway: 10.2.23.1
DNS Server: 10.10.30.10
Start IP: 10.2.23.11
Subnet Mask: 255.255.255.0
Maximum Users: 240
TFTP Server: 10.10.30.10

SCOPE: VOICE_F4 (VLAN 24)

Pool Name: VOICE_F4
Default Gateway: 10.2.24.1
DNS Server: 10.10.30.10
Start IP: 10.2.24.11
Subnet Mask: 255.255.255.0
Maximum Users: 240
TFTP Server: 10.10.30.10

SCOPE: VOICE_F5 (VLAN 25)

Pool Name: VOICE_F5
Default Gateway: 10.2.25.1
DNS Server: 10.10.30.10
Start IP: 10.2.25.11
Subnet Mask: 255.255.255.0
Maximum Users: 240
TFTP Server: 10.10.30.10

SCOPE: PRINTERS (VLAN 50)

Pool Name: PRINTERS
Default Gateway: 10.10.50.1
DNS Server: 10.10.30.10

```

Start IP:      10.10.50.11
Subnet Mask:   255.255.255.224
Maximum Users: 18

SCOPE: WIFI_CORP (VLAN 60)
Pool Name:     WIFI_CORP
Default Gateway: 10.20.60.1
DNS Server:    10.10.30.10
Start IP:      10.20.60.11
Subnet Mask:   255.255.254.0
Maximum Users: 488

SCOPE: WIFI_GUEST (VLAN 70)
Pool Name:     WIFI_GUEST
Default Gateway: 10.20.70.1
DNS Server:    8.8.8.8
Start IP:      10.20.70.11
Subnet Mask:   255.255.255.0
Maximum Users: 240

=====

=====
NOT in DHCP – STATIC ONLY:
  VLAN 30 SERVERS – manually configure each server's NIC
  VLAN 40 MGMT    – switch/router SVIs already in their configs
=====

=====

PT GUI checklist:
1. Server-1 NIC: IP 10.10.30.10, mask 255.255.255.192,
   gateway 10.10.30.1, DNS 10.10.30.10 (self)
2. Server-2 NIC: IP 10.10.30.20, mask 255.255.255.192,
   gateway 10.10.30.1, DNS 10.10.30.10
3. On Server-2: Services > DHCP > Service: On
4. Add each scope above on Server-2, click Save after each
5. Verify: from a PC on each VLAN, set NIC to DHCP → renew → IP appears
=====

```

Note on ACC-F2-1, ACC-F3-1, ACC-F4-1, ACC-F5-1: these access switches are identical in structure to ACC-GF-1 with the following per-floor substitutions: (a) hostname becomes ACC-Fn-1; (b) the local SVI on VLAN 40 changes its IP within 10.10.40.0/26 per Appendix A.4; (c) the data VLAN on user access ports changes to 12/13/14/15 (per floor); (d) the voice VLAN changes to 22/23/24/25 (per floor); (e) the trunk allowed-VLAN list changes to that floor's pair (e.g., F2: 12, 22, 40, 50, 60, 70, 100); (f) VLAN 30 server access ports are absent on F2 through F5 (the corresponding ports are reassigned as user-access ports in those floors' configurations).

Appendix D — Verification Command Reference

Quick-reference list of the commands used during testing. Detailed pass/fail results are recorded in Section 8.2 after testing.

D.1 Layer-1 / Layer-2

show ip interface brief	! interface status, every device
show interfaces trunk	! native VLAN and allowed list
show vlan brief	! local VLAN database
show spanning-tree vlan <n>	! root, forwarding, and blocking ports
show spanning-tree root	! root bridge per VLAN
show mac address-table	! L2 forwarding entries
show etherchannel summary	! Po1 state on both cores

D.2 Layer-3 and HSRP

show ip interface brief	! SVI status on cores
show standby brief	! HSRP per-group state
show standby vlan <n>	! detailed HSRP per VLAN
show ip route	! routing table
show ip route ospf	! OSPF-learned routes only
show ip ospf neighbor	! adjacencies (must show FULL)
show ip ospf interface brief	! OSPF interface state, cost

D.3 Services (DHCP, DNS, NTP, Syslog, NAT)

```
! DHCP (from a host)
ipconfig /all                                ! Windows host
ipconfig /release && ipconfig /renew
! DHCP relay (from the core)
show running-config interface vlan <n> ! confirm ip helper-address

! NTP
show ntp status                                ! must say "Clock is synchronized"
show ntp associations                          ! asterisk on the elected master

! NAT
show ip nat translations
show ip nat statistics

! Syslog/DNS – verified at the server via the PT services view
```

D.4 Failover Tests

```
! HSRP failover
CORE-1# config terminal
CORE-1(config)# interface vlan 11
CORE-1(config-if)# shutdown
! On a host in VLAN 11: continuous ping to 10.1.11.1 should resume in < 10 s
CORE-1(config-if)# no shutdown          ! restore

! ISP failover
EDGE-RTR# config terminal
```

```
EDGE-RTR(config)# interface g0/0/0
EDGE-RTR(config-if)# shutdown
! From a user PC: continuous ping to 1.1.1.1 should resume via ISP-B
EDGE-RTR(config-if)# no shutdown    ! restore
```